Creating a Dynamic DNS Server Using Twisted Python

This article discusses an event-driven networking framework called Twisted Python, with which one can build a Web server, email server, DNS server, IRC bot and client, as well as server protocols.

An event-driven program is a programming paradigm in which the flow of the program is determined by various events (user actions, sensor outputs or messages). This is in contrast to other programming paradigms, the most simple probably being a single-threaded synchronous program.

If your program is executed from top to bottom, and if you have an expensive operation like a database query or a network request, then one option is for this program to wait until it gets a response. This may not be time-efficient, but is certainly an easy option. The second option is to spit out this task into different threads of control, which will be managed by the operating system; and hence we get some progress. But this thread operation may be inconsistent in the case of file writing or value updating. In that case, you have to implement some locking mechanism to make sure data is not corrupted. The next option is the event-driven program, which involves single-threaded execution.

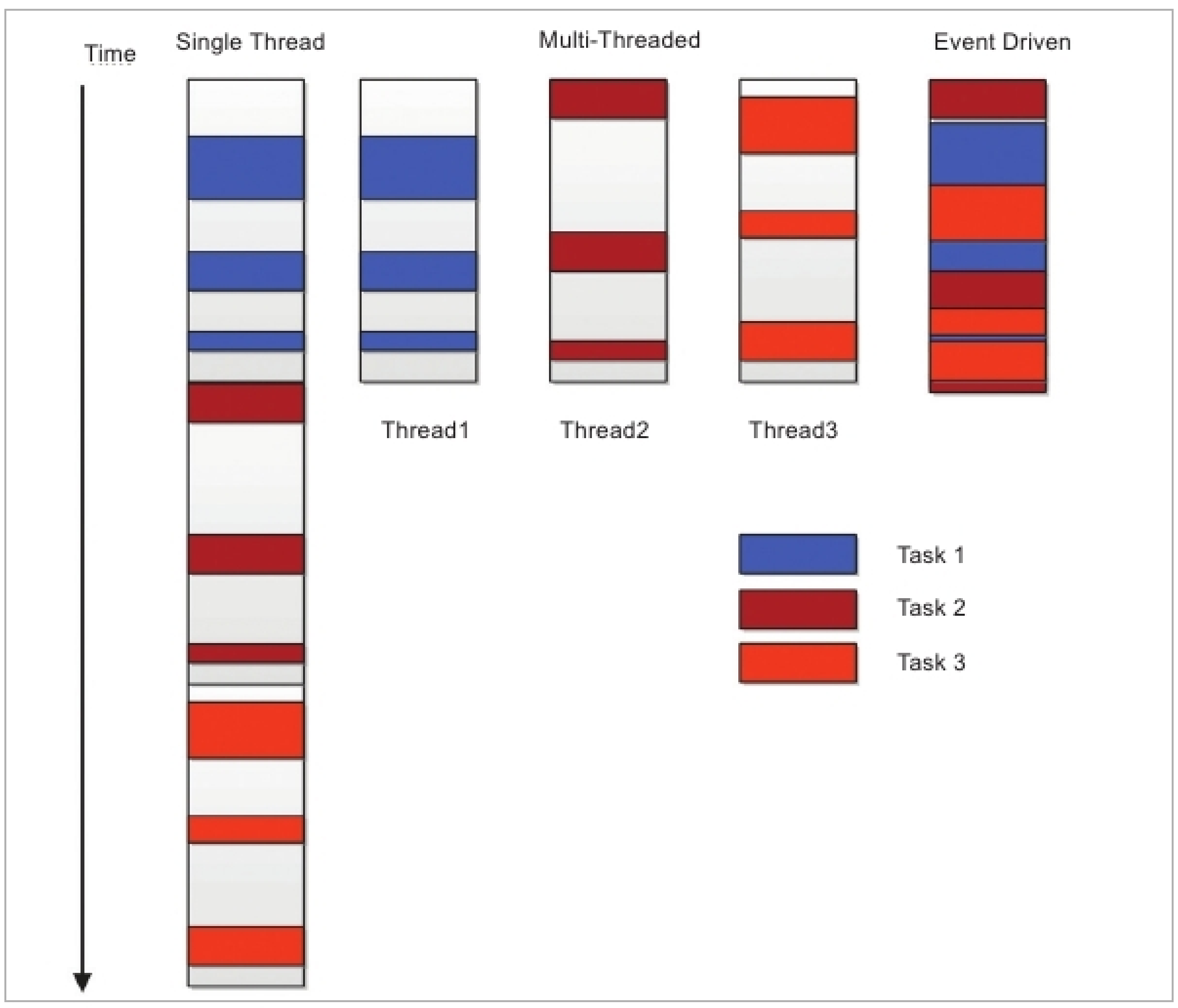

Figure 1: The difference between single-thread, multi-threaded and event-driven programs

An event loop accompanies event-driven programming

An event loop runs the following tasks in a continuous loop:

- Event detection
- Event handler triggering

Whenever the event loop is run, it detects which event has happened. Then, the event loop must determine the event callback and invoke it. The event loop is one thread running inside one process, which means that when an event happens, the event handler runs without interruption. A good choice for an event-driven program is when there are several tasks that can happen, probably independently.

Let us explore some components of Twisted Python.

Reactor: If you have an event-driven model, you need some kind of event loop that handles the listening to and dispatching of events for processing. That event loop is called the reactor in Twisted. The latter handles several kinds of events, such as network events, file events and timers.

Transport: Transport represents the connection between two endpoints communicating over a network. When